Nokia 1830 PSS Integrated OTDR

The 1830 PSS Integrated optical time-domain reflectometer (OTDR) unit simplifies the operations and maintenance for fiber optics networks by providing real-time monitoring of fiber spans, fault detection and localization. When fiber cuts occur, it is critical to quickly diagnose, locate, and repair the cable so that critical services are promptly restored. The Nokia integrated OTDR eliminates cost and time to schedule manual OTDR measurements.

The 1830 PSS automatically detects outages, scans the fiber, and notifies the network management center of the location of the fiber cut so that repair crews can be immediately dispatched, saving days in the repair cycle.



Benefits

- Reduces cost and simplifies network operations by real-time monitoring of fiber spans
- Eliminates time and cost of scheduling manual OTDR measurements
- Identifies faults within minutes so that repair crews can be dispatched to the exact fault location
- Reduces days or weeks of downtime
- Simplifies network turn-up with Raman amplifiers

Applications

- Automated network intelligence
  - Fault detection and localization
  - Monitoring real-time network performance
  - Optical intrusion detection
  - Simplified Raman provisioning
  - Correlation to baseline to monitor aging
- Single integrated unit supporting up to eight degrees

Product description

The Nokia 1830 PSS Integrated OTDR unit provides real-time monitoring of the fiber span, enabling fault detection and localization, as well as simplified network provisioning with Raman turn-up wizards. An industry standard OTDR signal is transmitted over the fiber and reflections are analyzed to determine event types, including connectors, fiber splices, and fiber breaks, along with the location of each event.

The OTDR unit supports eight optical inputs using LC connectors. The optical inputs can be connected to OTDR Tx or Rcv monitor ports on amplifier and iROADM cards.

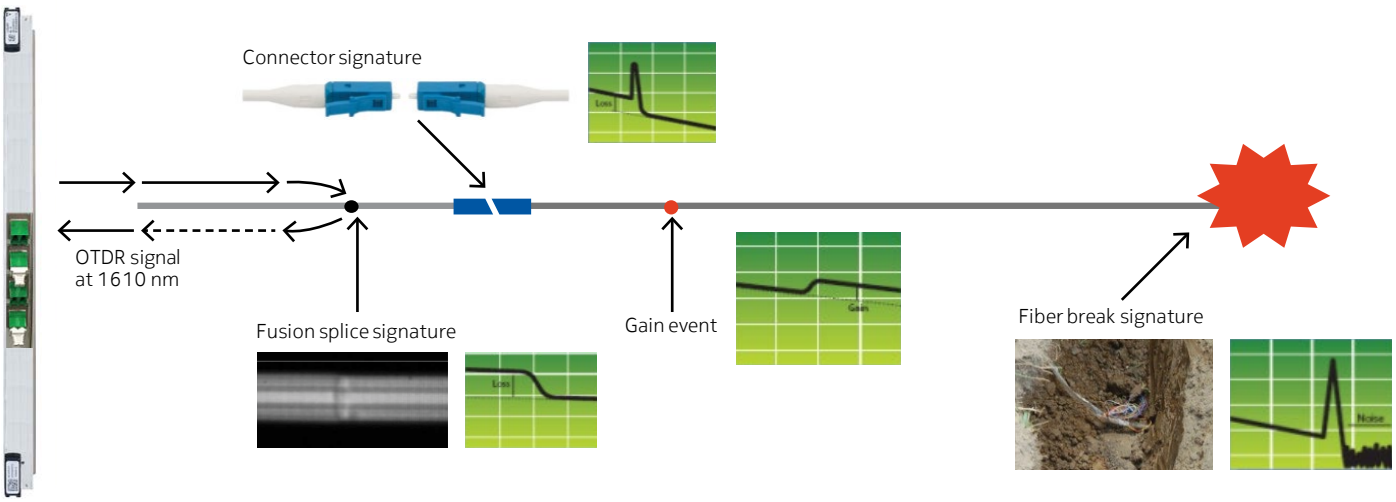
For amplifiers without integrated OTDR monitor ports, a MON-OTDR filter card is available to filter and insert the 1610 nm OTDR signal onto the line.

A standard 1610 nm wavelength is used for normal OTDR applications on C-band amplifiers. A wideband OTDR option is also available for use with the Nokia Ultra-Wideband (C+L) amplifiers and operates at 1625nm.

Nokia supported products

The OTDR unit is supported on Nokia 1830 PSS-32, 16ii, -8 and PSI-L systems

Figure 1. OTDR application with typical reflections from fiber events



OTDR versions

Unit Name	Part#	Description
OTDR	8DG62504AA	C-Band OTDR, high performance
OTDRWB	8DG63068AA	Wideband-OTDR for use on C+L ultra-wideband ROADM
OTDRM	8DG64184AB	C-Band OTDR, metro optimized performance
MON-OTDR	8DG63264AA	OTDR filter for use on amps without integrated OTDR ports

Technical specifications

Specifications			
Unit name	OTDR	OTDRWB (wideband)	OTDRM (metro)
Ports	8 OTDR input ports	8 OTDR input ports	8 OTDR input ports
OTDR wavelength	1610 nm	1625 nm	1610 nm
Range	> 120 km	> 120 km	> 50 km
Pulse widths	5 ns, 10 ns, 30 ns, 100 ns, 300 ns, 1 us, 3 us, 10 us, 20 us	5 ns, 10 ns, 30 ns, 100 ns, 300 ns, 1 us, 3 us, 10 us, 20 us	30 ns, 100 ns, 300 ns, 1 us, 3 us
Dynamic range	36 dB	36 dB	16 dB
Power consumption	29 W (max.)	30 W (max.)	30 W (max.)
Launch fiber	30 m integrated into OTDR		
Operating environment	Normal Humidity	5°C to 40°C (41°F to 104°F) 5% to 85%	
Physical	1-slot, full height		

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Document code: (June) CID200367